DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTECPE603B - BIG DATA ANALYTICS

Time : 3 Hours] **[Total Marks : 60**

Instructions to the Students:

1.Each question carries 12 marks.
2.Question No. 1 will be compulsory and include objective-type questions.
3.Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4.Use of non-programmable scientific calculators is allowed.
5.Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question) 12

1 **Which of the following is not a type of digital data?**
A) Structured data
B) Unstructured data
C) Semi-structured data
D) Discrete data

2 **Hadoop was originally developed by:**
A) IBM
B) Google
C) Apache Software Foundation
D) Yahoo!

3 **What is the main advantage of using Hadoop for Big Data processing?**
A) It requires expensive hardware
B) It can only process structured data
C) It allows distributed storage and parallel processing
D) It runs only on Windows OS

4 **Which of the following tools is used for importing data from relational databases into HDFS?**
A) Flume
B) Hive
C) Sqoop
D) Pig

5 **What is the main purpose of Hadoop Archives (HAR files)?**
A) To compress files for transmission
B) To allow real-time analytics
C) To store large numbers of small files efficiently
D) To reduce Namenode memory usage

6 **In Hadoop, which of the following is used for streaming log data into HDFS in real time?**
A) Sqoop
B) Pig
C) Flume
D) Hive

7 **In a MapReduce job, what is the main purpose of the shuffle and sort phase?**
A) To combine multiple jobs
B) To distribute data across clusters
C) To transfer map output to reducers and sort it by key
D) To delete temporary files

8 **What happens if a MapReduce task fails during execution?**

A) The entire job is aborted immediately
B) It is automatically re-executed on another node
C) Hadoop ignores the failed task
D) The user is prompted to restart the task manually

9 **In Hive, what is the role of the Metastore?**

A) It executes Hive queries
B) It stores the actual data files
C) It manages metadata about tables, partitions, and schemas
D) It loads data into HDFS

10 **Which of the following is the scripting language used in Apache Pig?**

A) SQL
B) Pig Latin
C) HiveQL
D) Grunt

11 **In Apache Kafka, which component is responsible for receiving data from producers and storing it?**

A) Zookeeper
B) Broker
C) Consumer
D) Topic

12 **In Apache Spark, what is the function of the Driver Program?**

A) It stores all the data
B) It manages cluster resources
C) It coordinates tasks and maintains the SparkContext
D) It serves as a backup node

Q2. Solve the following.

A) Explain the different types of digital data. How does the variety of data influence Big Data analytics? 6

B) Describe the history and evolution of Hadoop. How has Apache Hadoop transformed big data processing? 6

Q3. Solve the following.

A) Explain the design and key concepts of HDFS. What are its core components, and how does it ensure fault tolerance and high availability? 6

B) Describe the data ingest methods in Hadoop using Flume and Sqoop. Compare their purposes and use cases. 6

Q4. Solve Any Two of the following.

A) What are the common types of failures in a MapReduce job, and how does Hadoop handle them? 6

B) What are the different MapReduce data types and formats supported in Hadoop? Explain with examples. 6

C) What are the features of a Hadoop cluster? Describe the components and their roles in executing MapReduce jobs. 6

Q5. Solve Any Two of the following.

A) Differentiate between the local mode and MapReduce mode of Pig execution. In which scenarios would each be used? 6

B) Write a short note on Hive Shell and HiveQL. How does Hive differ from traditional RDBMS? 6

C) What is Big SQL? Describe its significance and how it enables SQL queries on Hadoop data. 6

Q6. Solve Any Two of the following.

A) Explain the architecture of Apache Kafka with a neat diagram. Describe the roles of its key components: Producer, Broker, Consumer, Topic, and Zookeeper. 6

B) Describe the architecture of Apache Spark. Explain the roles of the Driver Program, Cluster Manager, and Executors. 6

C) What is Kerberos authentication? Explain its key components and how they work together to provide secure access. 6

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